名称	序号	类型[1]	功能
GND	1	P	接地
3V3	2	P	供电
EN	3	I	高电平：芯片使能； 低电平：芯片关闭； 注意：未悬空 EN 管脚浮空。
SENSOR_VP	4	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_VN	5	I	GPIO39, ADC1_CH3, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P (32.768 kHz 晶振输入), ADC1_CH4, TOUCH9, RTC_GPIO9
IO33	9	I/O	GPIO33, XTAL_32K_N (32.768 kHz 晶振输出), ADC1_CH5, TOUCH8, RTC_GPIO8
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	接地
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
NC	17	-	请无名脚下方说明[2]
NC	18	-	请无名脚下方说明[2]
NC	19	-	请无名脚下方说明[2]
NC	20	-	请无名脚下方说明[2]
NC	21	-	请无名脚下方说明[2]
NC	22	-	请无名脚下方说明[2]
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTDO, HSPICS0, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPIWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPIHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16[7]	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICS0, HS1_DATA6, EMAC_RX_CLK
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7

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表 2-模组管脚

名称	序号	类型[1]	功能
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	-	-
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPIWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	接地

[1] P：电源；I：输入；O：输出。
[2] ESP32-DOWD-V3/ESP32-DOWDRQ-V3 芯片上的 GPIO6 至 GPIO11 用于连接模组上集成的 SPI flash。不再启动更相容管脚。
[3] 由带有 QSPI PSRAM（即内置芯片为 ESP32-DOWDRQ-V3）的模组中，IO16 用于连接至嵌入式 PSRAM。不可用于其他用途。

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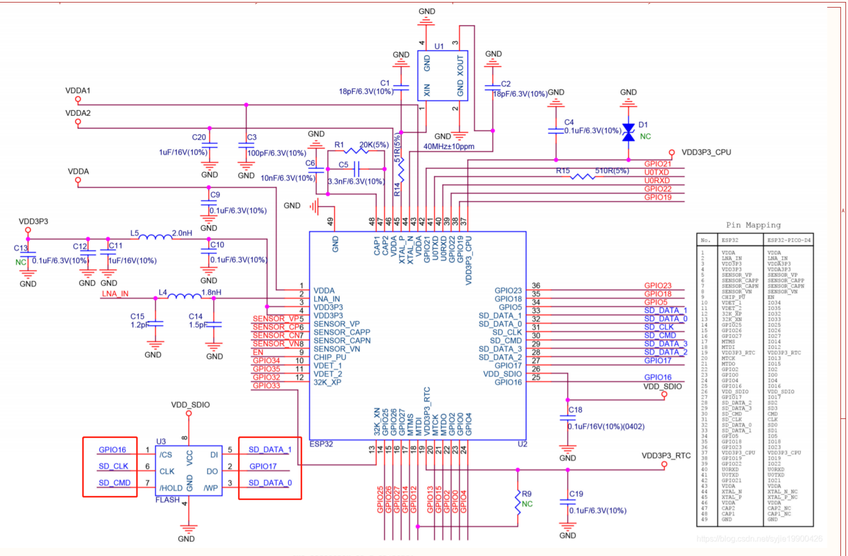
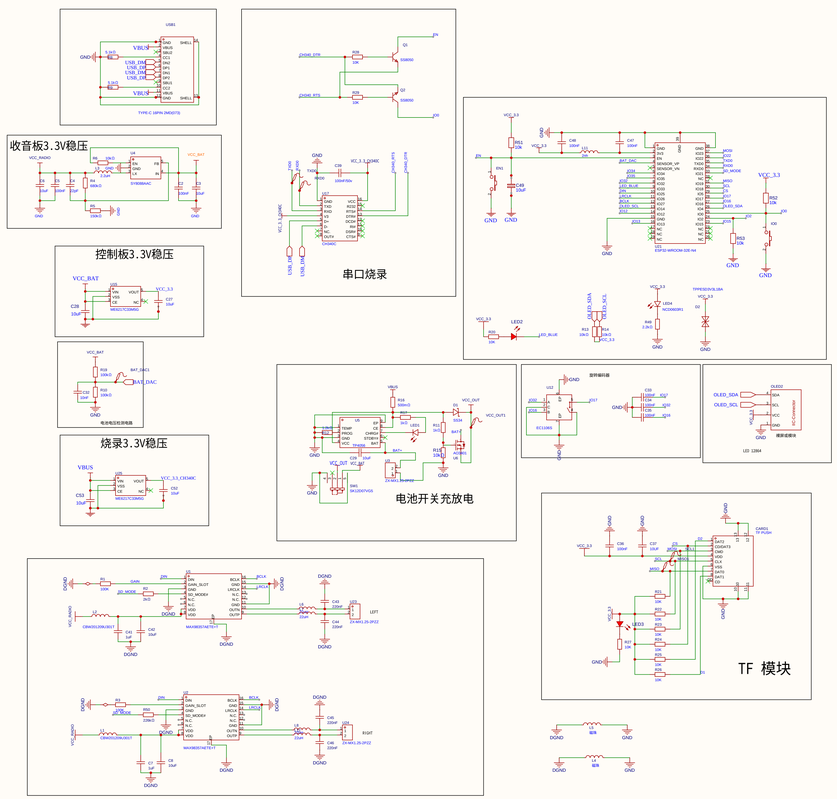
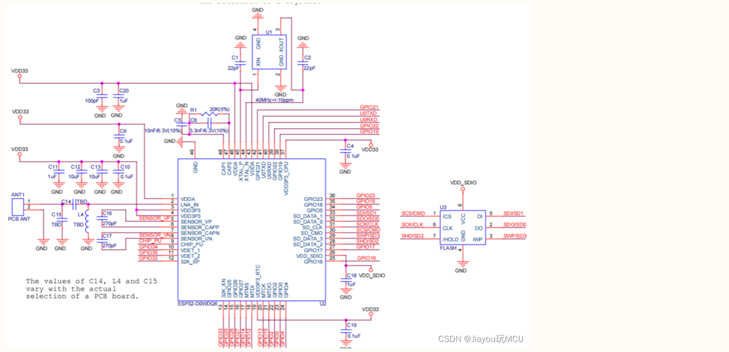
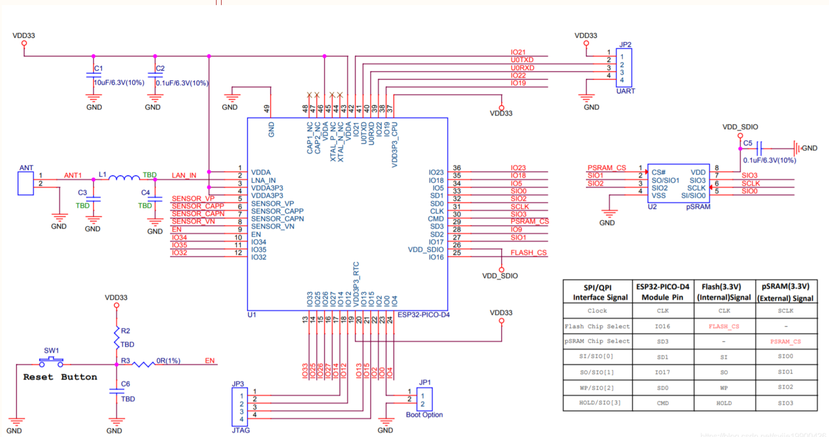








SPI/QPI Interface Signal	ESP32-PICO-D4 Module Pin	Flash(3.3V) (Internal)Signal	pSRAM(3.3V) (External) Signal
Clock	CLK	CLK	SCLK
Flash Chip Select	IO16	FLASH_CS	-
pSRAM Chip Select	SD3	-	PSRAM_CS
SI/SIO[0]	SD1	SI	SIO0
SO/SIO[1]	IO17	SO	SIO1
WP/SIO[2]	SD0	WP	SIO2
HOLD/SIO[3]	CMD	HOLD	SIO3

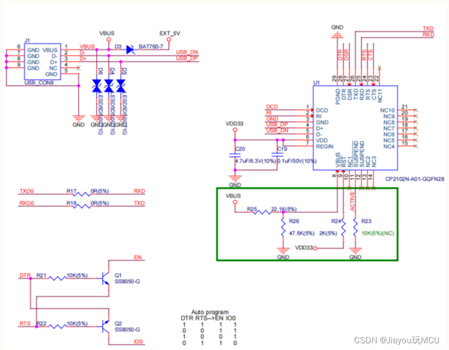
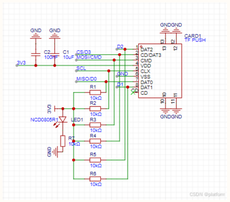


表 8.增益选择

GAIN_SLOT	IS/LJ 增益 (dB)
通过 100kΩ 连接到 GND ±5% 电阻	15
连接到 GND	12
无关	9
连接到 V	6
连接到 Vthrough 100kΩ ±5% 电阻	3



原理图	Schematic1	创建日期	2026-02-22
		更新日期	2026-04-10
板子	Board1	图页	P1
绘制		网络流媒体播放器	
审阅			
嘉立创EDA	版本 V1.0	尺寸 A4	页 1 共 1
		嘉立创EDA	